

Comprehensive Guide to Potato Plant Health and Early Blight Management

1. Overview

Potatoes are one of the most important and widely grown vegetable crops in the entire world. In fact, potatoes are the fourth largest crop grown globally, coming in just behind wheat, rice, and corn. In the United States alone, the average person eats about 130 pounds of potatoes every single year. Because they are so popular and provide so much food for people everywhere, keeping potato plants healthy is a very big deal for both small backyard gardeners and massive commercial farmers.

However, growing potatoes is not always easy. From the moment the potato seed is planted in the dirt to the day the mature potatoes are dug up, the plants face a wide variety of threats. While insects and tiny worms in the soil can cause problems, fungal diseases are often the biggest threat to a healthy potato crop. Many people have heard of "late blight," which is the highly destructive disease that caused the famous Irish Potato Famine in the 1840s. But there is another, very common disease that growers must fight every single year: early blight.

Early blight is a fungal disease that attacks plants in the nightshade family, which includes potatoes, tomatoes, eggplants, and peppers. The name "early blight" can be a little bit confusing. It does not necessarily mean that the disease only happens early in the spring. Instead, it means that the disease usually shows up earlier in the plant's life cycle than late blight does, often starting in the middle of the summer when the potato plants are blooming and beginning to form their tubers underground.

When a potato plant is perfectly healthy, its leaves are a vibrant, solid green color. The stems are strong, and the plant uses the energy from the sun to grow large, healthy potatoes hidden in the soil. But when a plant is attacked by early blight, the fungus slowly destroys the green leaves. Because the plant uses its leaves to make food from sunlight, losing these leaves means the plant literally starves. The plant becomes too weak to grow large potatoes, which means the farmer or gardener will get a much smaller harvest. Even worse, the fungus can sometimes travel down into the soil and attack the potatoes themselves, causing them to rot and become completely useless for eating or selling.

Managing early blight is not just about spraying chemicals when you see a problem. True plant health requires a combination of good habits. It means choosing the right kind of potatoes to plant, giving them the right amount of space, watering them properly, keeping the soil healthy, and knowing exactly what to look for so you can stop the disease before it spreads too far. This guide will walk you through every single step in simple, easy-to-understand language. We will cover how to spot the disease, what causes it to spread, how to treat it naturally and chemically, and how to prevent it from ever coming back into your garden or farm.

2. Symptoms

If you want to keep your potato plants healthy, you need to know exactly what a sick plant looks like. Catching the disease early is the absolute best way to save your crop. Early blight shows up in very specific ways, and it changes as the season goes on. Here are the visible signs of the

disease, broken down by which part of the plant is affected.

Symptoms on the Leaves

The leaves are almost always the very first place you will see early blight. Because the fungus loves moisture and humidity, it usually starts on the oldest leaves that are closest to the damp soil.

- **Small Dark Spots:** The very first sign of trouble will be tiny, dark spots on the lower leaves. These spots are usually dark brown or black. When they first appear, they might be incredibly small, sometimes as tiny as the head of a pin, but they can quickly grow to about 1/8 of an inch wide.
- **The Bullseye Pattern:** As the dark spots get bigger, they run into the tiny veins inside the leaf. Because the veins are tough, they force the spots to take on irregular, angular shapes. Inside these dark, angular spots, you will see something very special: rings. The spots will have dark and light brown circles inside them that look exactly like a target or a bullseye. This "target-like" pattern is the most important clue. If you see the bullseye, you almost certainly have early blight.
- **Yellowing Leaves (Chlorosis):** The plant does not just sit there while the fungus attacks; the leaf tissue starts to break down. Around the edges of the dark bullseye spots, the leaf will turn a sickly yellow color. This yellowing is called chlorosis. As the spots grow and multiply, the yellow color spreads until the entire leaf loses its healthy green color.
- **Drying and Dying:** Eventually, the heavily infected leaves will completely dry up, wither, and die. But unlike the leaves of some trees that fall to the ground when they die, potato leaves killed by early blight usually stay firmly attached to the stem, hanging there like dry, brown paper.
- **Moving Up the Plant:** If you do not treat the disease, it will not stay on the bottom leaves. It will slowly climb higher and higher up the plant. By the end of the summer, the top leaves will be peppered with hundreds of small dark spots that merge together to form massive dead areas.

Symptoms on the Stems

While the leaves take the hardest hit, the stems of the potato plant are not safe from the fungus.

- **Dark Streaks:** You may notice long, dark brown or black streaks running up and down the main stems of the plant.
- **Petiole Lesions:** You might also see these dark spots on the petioles. The petiole is the small, thin stalk that connects the flat leaf to the main stem. When the petiole gets heavily infected, the leaf attached to it will eventually die because it can no longer receive water from the main plant.

Symptoms on the Tubers (The Potatoes)

The ultimate goal of growing potatoes is to harvest the tubers under the ground. Unfortunately, early blight can ruin them.

- **Sunken Dark Spots:** If the fungus reaches the potatoes in the soil, it will cause dark spots on the skin. These spots are usually round or irregular, and they look pushed in or sunken, almost like a small crater.
- **Raised Purple Borders:** Around the edge of the sunken crater, the potato skin might look

slightly raised and turn a dark purple or dark brown color.

- **Corky Texture:** If you take a knife and cut into one of these dark spots, you will see that the inside of the potato is ruined. The flesh directly under the spot will be dry, dark brown, and have a tough, leathery, or corky texture.
- **Storage Issues:** Potatoes with early blight spots are usually not fit to eat fresh, and factories that make potato chips or french fries have to peel a lot of the potato away to get rid of the dark, corky tissue. Interestingly, the early blight fungus does not spread from one potato to another while they are sitting in a storage bin. However, the sunken spots break the potato's skin, which lets all the moisture leak out. This causes the infected potatoes to shrivel up and become wrinkled during storage.

How to Tell the Difference Between Early Blight and Late Blight

It is very important to know which disease you have, because they act differently.

- **Early Blight:** Has the famous "bullseye" target pattern inside the brown spots. It starts on the bottom leaves and moves up slowly.
- **Late Blight:** Does not have bullseye rings. Instead, it causes large, messy, pale green or brown patches that cross right over the leaf veins. More importantly, late blight causes a fuzzy, white mold to grow on the underside of the leaves when the weather is wet. Late blight also kills plants incredibly fast, sometimes destroying a whole field in just a few days, whereas early blight is a slower, steadier disease.

3. Causes & Spread

To fight early blight effectively, you need to understand what causes it, how it survives, and how it travels from one plant to another.

The Pathogen: *Alternaria solani*

Early blight is not caused by a bug or a worm. It is caused by a microscopic fungus with the scientific name *Alternaria solani*. In simple terms, a fungus is a type of living organism that absorbs food from the things it grows on. *Alternaria solani* is a specific type of fungus that happens to love eating the tissues of potato and tomato plants.

This fungus is incredibly tough and highly adaptable. Scientists call it a "polycyclic" pathogen. In plain language, this means the fungus does not just attack once a year. It can go through its entire life cycle—infesting a leaf, growing, and producing babies—multiple times in a single summer. Because it repeats this cycle over and over again, a small infection can explode into a massive problem very quickly.

To reproduce, the fungus does not lay eggs. Instead, it creates thousands of tiny, invisible microscopic particles called spores. You can think of these spores as the seeds of the fungus. These spores are dark in color, which helps protect them from the sun's ultraviolet rays. This makes them very strong and able to survive tough weather conditions.

How the Disease Spreads

If you have a perfectly healthy potato plant, how does the *Alternaria solani* fungus actually get to it? There are several ways this sneaky pathogen travels around your garden or farm.

- **Overwintering in Plant Trash:** The most common way early blight starts is from the ghost of last year's garden. When winter comes, the fungus does not die. Instead, it goes to sleep inside the dead potato leaves, vines, and stems that are left lying on the dirt. When spring arrives and you plant new potatoes in that same dirt, the fungus wakes up and immediately attacks the young, vulnerable plants.
- **Splashing Rain and Watering:** Once the fungus is in the dirt, it needs a way to get up onto the leaves. When heavy raindrops hit the bare ground, they splash tiny drops of muddy water into the air. If the dirt is infected, those muddy water drops carry the fungal spores right onto the bottom leaves of your potato plant. This is why the disease almost always starts on the lowest leaves.
- **Wind Travel:** When the fungus successfully attacks a leaf, it grows a new crop of spores. Because these spores are microscopic and incredibly light, a strong breeze can easily blow them off the infected plant. The wind can carry these spores through the air to healthy plants nearby, or even to a neighbor's garden.
- **Weeds that Hide the Fungus:** Potatoes and tomatoes are not the only plants that get early blight. There are several wild weeds, especially "hairy nightshade" and "black nightshade," that the fungus loves to eat. If you let these weeds grow near your garden, the fungus will live on them, multiply, and constantly send spores blowing toward your potato plants.
- **Planting Sick Potatoes:** Sometimes, the disease starts because the gardener actually planted it. If you save potatoes from a sick plant to use as seed potatoes for the next year, you are burying the disease right in the ground.

Environmental Triggers (The Perfect Storm)

Even if the fungus is present, it needs specific weather conditions to cause a major outbreak. Knowing what weather the fungus likes helps you know when to be most careful.

- **Warm and Wet Weather:** The early blight fungus absolutely loves moisture. If the weather is warm and there is a lot of rain, heavy morning dew, or high humidity, the spores will wake up and stick to the wet leaves.
- **Alternating Weather Cycles:** Interestingly, the disease spreads fastest when the weather keeps changing. A cool, cloudy, wet morning allows the fungus to attach to the leaf. Then, when the afternoon turns warm, sunny, and dry, the fungus is stimulated to release thousands of new spores into the wind.
- **Stressed, Aging Plants:** The early blight fungus is somewhat of an opportunist. It usually does not attack strong, young, actively growing plants. Instead, it waits until the potato plants start to get older and begin producing their flowers and tubers. It also loves to attack plants that are stressed. If your plants are starving because the soil doesn't have enough nutrients, or if they are suffering from damage, their immune systems are weak. A weak plant is an easy target for early blight.

4. Treatment / Remedies

When early blight shows up on your potato plants, it is very hard to make it go away completely. However, you can absolutely slow it down and protect the healthy green leaves that are still left. Depending on your preference and the size of your garden or farm, there are many natural,

organic, and chemical ways to fight back against the fungus.

Natural / Home Remedies and Biopesticides

Many home gardeners and organic farmers do not want to use harsh synthetic chemicals on their food. Luckily, modern science has discovered a huge variety of natural tools—often called biopesticides—that use nature's own defenses to stop early blight. These natural products are usually certified for organic farming and use good bacteria, good fungi, or essential oils.

Good Bacteria and Fungi: Just like humans take probiotics for good gut health, plants can be protected by good microbes that fight off the bad ones.

- **Bacillus subtilis:** This is a very helpful type of natural bacteria. Products like Serenade ASO, Aviv, Companion, and Theia use this bacteria. When you spray this on your potato leaves, it acts like a vaccine. It triggers the plant's natural immune system to wake up and fight off the early blight fungus. It also creates a protective layer on the leaf that the fungus cannot break through.
- **Bacillus amyloliquefaciens:** Found in products like Double Nickel 55, Serifel, and Stargus, this beneficial bacteria naturally produces chemicals that stop the early blight fungus from growing. It is often used to suppress the disease from getting worse.
- **Bacillus mycoides:** Used in a product called LifeGard WG, this microbe acts as a biological plant activator. You spray it on the plant, and it tells the potato plant to build up its internal defenses against the disease.
- **Streptomyces lydicus:** This is a soil bacteria found in the product Actinovate AG. When sprayed on the leaves along with a natural sticker so it doesn't wash off, it simply crowds out the early blight fungus so it has no room to grow.
- **Gliocladium catenulatum:** This is actually a good fungus found in a product called Prestop. When you spray it on the plant, this good fungus actually attacks and parasitizes the bad early blight fungus.

Essential Oils and Botanical Extracts: Plants have developed their own natural oils to fight off diseases for millions of years. We can extract these oils and spray them on our potatoes.

- **Essential Oil Blends:** Products like GreenFurrow BacStop, GreenFurrow EF 400, and Sporan EC2 use highly concentrated mixes of natural oils like clove, rosemary, peppermint, cottonseed, thyme, garlic, and cinnamon. When sprayed on the plant, these strong smelling oils physically destroy the outer shell of the fungal spores.
- **Tea Tree Oil:** The product Timorex Act contains tea tree oil, which is famous for its natural anti-fungal properties.
- **Neem Oil Extract:** A product called Trilogy uses a special, clarified extract from the neem tree. When sprayed, it coats the potato leaves with a natural barrier that stops the fungus from taking hold.
- **Paraffinic Oil:** Organic JMS Stylet-oil is a highly refined oil that literally coats and suffocates the early blight spores, making it impossible for them to breathe or stick to the potato leaf.

Mineral Sprays (Organic): There are also natural minerals that change the environment on the leaf so the fungus cannot survive.

- **Potassium Bicarbonate:** Found in Carb-O-Nator, this is a very close relative of common household baking soda. When you spray it on the leaves, it makes the surface too alkaline (the opposite of acidic) for the fungus to grow.
- **Hydrogen Dioxide:** Products like OxiDate 2 and PerCarb use forms of hydrogen peroxide to instantly oxidize and kill the fungus on contact. The best part is that it breaks

down into simple water and oxygen, leaving no toxic residue behind.

- **Copper Sprays:** Copper has been used by farmers for centuries. Organic copper sprays like Bonide Liquid Copper, Champ Formula 2, Cueva, Cuprofix Ultra 40, Kocide, Liqui-Cop, Nu-Cop 50, and Previsto act as a shield. When the fungal spore lands on a leaf covered in copper, the copper destroys the spore before it can infect the plant. However, you must use copper carefully, because spraying too much year after year can cause toxic levels of copper to build up in your garden dirt.

Physical Action (Pruning and Cleaning): Sometimes the best remedy is simply removing the problem by hand.

- **Plucking Sick Leaves:** If you check your plants often and spot the first few lower leaves showing the dark bullseye spots, immediately pinch or cut those leaves off the plant. Getting the sick leaf out of the garden physically removes thousands of spores. It is perfectly safe to remove up to one-third of the plant's leaves if you catch the disease early. Never remove more than a third, or the plant will starve.
- **Proper Disposal:** Never just throw the sick leaves on the ground. You must burn them, bury them very deep, or put them in a sealed bag in your household garbage.
- **Sanitize Tools:** If you use pruning shears to cut off a sick leaf, the tool is now covered in invisible spores. If you then cut a healthy plant, you will infect it. Always wash your hands and clean your tools with a simple mixture of water and household bleach (a 0.5% sodium hypochlorite solution is perfect) before touching healthy plants.

Chemical Remedies

For very large commercial farms, or in areas where the weather makes early blight incredibly aggressive, natural remedies might not be strong enough. In these cases, farmers use synthetic chemical fungicides.

Because early blight only attacks older plants, spraying chemicals when the plants are tiny babies is a waste of money and time. A chemical spray program should usually start when the potato plants reach about 12 inches tall, or right when the very first spots are seen on the bottom leaves in late summer.

The Danger of Fungicide Resistance: Because the *Alternaria solani* fungus goes through its life cycle so fast, it is very good at mutating and getting used to chemicals. If a farmer sprays the exact same chemical all summer long, the weak fungi will die, but the strong ones will survive, breed, and create a "super-fungus" that the chemical can no longer kill.

To stop this from happening, scientists put different chemicals into groups called "FRAC groups" based on how they kill the fungus. A good farmer must practice "resistance management" by rotating these chemicals. You should never apply chemicals from the exact same FRAC group two times in a row.

Common Chemical Fungicides:

- **Multi-Site Inhibitors (Group M5):** These are chemicals like Chlorothalonil. They are sold under names like Bravo Ultrex, Bravo Weather Stik, Echo 720, Echo 90 DF, and Bonide Fung-onil for home gardeners. These are excellent because they attack the fungus in many different ways at once. Because the attack is so complex, it is almost impossible for the fungus to develop a resistance to it.
- **SDHI Inhibitors (Group 7):** These chemicals stop the fungus from breathing. They include products like Fontelis, Vertisan, Luna Tranquility, and Luna Pro. However, farmers must be very careful. One older Group 7 chemical called boscalid (sold as Endura) is no longer recommended in many places because the fungus has completely mutated and

become immune to it.

- **Group 9 and Group 28:** Other powerful chemicals include pyrimethanil (found in Scala) and propamocarb hydrochloride (found in Previcur Flex). These attack the fungus by stopping it from building strong cell walls.

When using chemicals, they are usually sprayed every 7 to 14 days to keep a protective shield on the growing leaves. It is also critical to read the label for the "Preharvest Interval" or PHI. This is the number of days you must wait between the final chemical spray and the day you dig up the potatoes to ensure they are safe to eat.

5. Preventive Measures

The absolute best way to manage early blight is to never let it get started in the first place.

Preventing the disease is much cheaper, easier, and less stressful than trying to cure a plant that is already sick. By changing a few simple habits in how you plant and care for your garden, you can make it incredibly difficult for the fungus to survive.

Crop Rotation

This is arguably the most important rule in all of gardening and farming. As mentioned earlier, the early blight fungus survives the cold winter by sleeping in the dead leaves and roots of last year's potato plants.

- **Wait Several Years:** If you plant potatoes in the exact same spot two years in a row, the sleeping fungus will wake up in the dirt and immediately attack your new plants. To stop this, you must use "crop rotation." This means you must plant your potatoes in a completely different area of the garden or farm. You should wait at least two, and ideally three to four full years, before planting potatoes (or tomatoes and peppers) in that original spot again.
- **Plant Non-Hosts:** While you wait those three to four years, you should plant "non-host" crops in that dirt. Plants like corn, small grains, wheat, or soybeans cannot catch early blight. With nothing to eat, the early blight fungus left in the dirt will eventually starve and die entirely.
- **Physical Distance:** Moving your potatoes 10 feet to the left is not enough, because the wind can blow the spores. If possible, try to plant this year's potatoes at least 225 to 450 yards away from where last year's potatoes grew. If you farm on a large scale, planting tall strips of wheat between your fields can act as a wall to block the wind from blowing spores around.

Choosing Resistant Varieties

Not all potatoes are created equal. Thanks to decades of hard work by scientists and plant breeders, there are many varieties of potatoes that have a naturally strong immune system against early blight. While no potato is 100% immune, choosing a "resistant" variety means the plant will suffer far less damage and produce a great harvest without needing heavy chemical sprays. When buying seed potatoes, look for the letters "EB" in the catalog, which stands for Early Blight resistant.

Highly Resistant Potatoes to Plant:

- **Sarpo Mira:** This is a famous potato variety loved by organic farmers. It is a champion at fighting off both early and late blight. It grows very well even when the weather is dry, and it produces a huge amount of large, uniform potatoes.
- **Defender:** Created by the USDA specifically to fight blight, this potato keeps its green

leaves and keeps growing strong long after other potato types have died from the fungus.

- **Blackberry:** Many gardeners consider this one of the most resistant varieties available. The leaves stay remarkably healthy, and it almost never suffers from dark, corky rot on the underground tubers.
- **Orla:** This is a fantastic, flexible potato. It strongly resists tuber blight. You can dig it up early in the summer for small, tender potatoes, or leave it in the ground until fall for a massive main crop harvest.
- **Setanta:** An excellent, heavy-producing potato that rarely needs chemical sprays because its natural defenses are so high.
- **Texla:** A perfectly round potato that is highly resistant to blight, though it does not like to grow in heavy, wet clay soil.
- **Other Great Choices:** There are many other varieties with wonderful resistance, including Desiree, Lehigh, Red Norland, Red Gold, Superior, Reba, Masquerade, Chieftain, Red Thumb, French Fingerling, Pinto, Baltic Rose, Blue Anneliese, Gatsby, and Heidi Red.

Potatoes with Intermediate Resistance: Some potatoes might get a little bit sick, but they are so strong that they produce a huge harvest anyway. These include Elba, Katahdin, German Butterball, Soraya, and Upstate Abundance. Conversely, if you live in a wet area, you should absolutely avoid planting highly susceptible potatoes like Norchip, because they will catch the disease very easily.

Proper Watering Practices

Because the fungus needs moisture to wake up and attack, how you water your plants is critical.

- **Stop Overhead Sprinklers:** Watering your garden from above with a watering can or a tall sprinkler is very bad for potatoes. It gets all the leaves soaking wet, and it splashes muddy, infected dirt up onto the bottom of the plant.
- **Use Drip Irrigation:** The best way to water is using a drip irrigation system or soaker hoses. These are tubes that lie flat on the dirt and slowly drip water right into the soil. This gives the roots all the water they need, but keeps the leaves completely bone dry so the fungus cannot grow. If you must water by hand, hold the hose very low to the ground and water the base of the plant carefully.
- **Timing Your Watering:** If you are a commercial farmer and must use giant overhead sprinklers, timing is everything. Never water in the evening or during cool, cloudy weather. If you water at night, the leaves stay wet for 10 hours in the dark, giving the spores plenty of time to infect the plant. Always water early in the morning on a sunny day. This ensures the sun will quickly dry the leaves off before nightfall.
- **Mulching the Soil:** To stop mud from splashing up onto the leaves when it rains, cover the bare dirt around your potato plants. You can lay down a thick layer of clean straw, dried leaves, landscaping fabric, or plastic mulch. This creates a physical blanket over the soil, trapping the fungus down in the dirt where it can't hurt the leaves.

Airflow and Spacing

Fungus loves still, damp air. To fight it, you need wind and sunshine.

- **Give Them Room:** When planting your seed potatoes, do not crowd them together. Make sure there is plenty of space between each plant. When there is space, the wind can blow freely through the leaves. Good airflow dries out the morning dew quickly, taking away the

moisture the fungus needs.

- **Weed the Garden:** Keep your garden completely free of weeds. Tall weeds act like walls that block the wind and trap humidity around your potato plants. Also, remember that weeds like hairy nightshade actually carry the disease, so pulling them out removes the fungus from the area.

Total Sanitation

A clean garden is a healthy garden.

- **Start Clean:** Always buy certified, pathogen-free seed potatoes from a trusted garden center. Never try to plant a potato from a sick plant, because you will just introduce the disease to the dirt.
- **Fall Cleanup:** When the harvest is over in the fall, your work is not done. Do not leave the dead potato vines rotting in the field. Commercial farmers should use large plows to flip the soil and bury the dead plant trash deep underground where the spores will safely rot away. Home gardeners should rake up all the dead vines and remove them entirely.

6. Suggestions & Best Practices

Beyond direct treatments and prevention, there are several advanced techniques and habits that experienced growers use to ensure a massive, healthy harvest.

Fertilization and Soil Nutrition

A hungry plant is a weak plant. The early blight fungus specifically targets potato plants that are starving for nutrients, especially when the plant puts all its energy into growing flowers and potatoes mid-summer.

- **Nitrogen is Key:** You must keep the plant vigorous and strong by ensuring it has enough nitrogen in the soil just as the flowers start to open. Nitrogen keeps the leaves green and functioning.
- **Balance Phosphorus and Potassium:** Make sure your soil has adequate phosphorus, as a lack of phosphorus makes the plant highly vulnerable to severe blight damage. However, do not go overboard. Adding far too much heavy potassium fertilizer can actually cause imbalances that hurt the plant. The best practice is to do a simple soil test before planting to know exactly what food your dirt needs.

Using Predictive Weather Models (For Serious Growers)

Instead of guessing when to spray protectants, modern technology can tell you exactly when the fungus is about to attack.

- Agricultural universities have created weather tracking apps, like the NDAWN Potato Blight app.
- These apps track the temperature, humidity, and how many hours the leaves have been wet. They calculate this data into something called "P-Days" or Physiological Days.
- Research shows that the very best time to spray your first protective treatment is exactly 200 P-Days after your potato plants first poke out of the soil in the spring. Waiting for this exact moment saves money on unnecessary sprays and hits the fungus right when it is

most vulnerable.

Composting Rules for Gardeners

Many gardeners love to make compost out of dead leaves and vegetable scraps to feed their soil. However, if you put potato leaves infected with early blight into a normal compost pile, the tough spores will survive, and you will accidentally spread the disease all over your garden the next year when you use the compost. You can only compost sick leaves if you follow very strict "hot composting" rules :

- **The Right Size:** Your compost bin must be between 3 to 5 feet wide and 3 to 5 feet high. If it is smaller, it won't get hot enough. If it is bigger, oxygen can't get to the center.
- **Building Layers:** You must build the pile carefully. Start with 8 to 10 inches of plant waste. Then add a layer of nitrogen, like livestock manure or half a cup of fertilizer. Then add a one-inch layer of soil. Keep stacking these layers until the pile is 5 feet tall, watering it so it feels like a damp sponge.
- **Turning and Heating:** You must use a pitchfork to turn the pile inside out once or twice a month to let oxygen in.
- **The Magic Temperature:** The microbes working in the pile will create immense heat. The center of the pile must reach between 130 and 160 degrees Fahrenheit. Only this intense heat will melt and kill the early blight spores. If you are not willing to do this much work, do not put sick leaves in your compost. Burn them or throw them in the trash instead.

Harvesting to Prevent Rot How you dig the potatoes up can decide whether they rot in storage or last all winter.

- **Wait for Full Maturity:** Never dig up potatoes while the green vines are still alive and actively growing. You must wait until the vines are completely dead and brown. Waiting allows the potato skin underground to become thick and tough.
- **Be Gentle:** When you finally dig them up, be extremely careful not to cut, scrape, or bruise the potatoes with your shovel. Any cut in the skin is an open door for fungi and bacteria in the dirt to enter the potato and rot it from the inside out.

7. References / Sources

The information compiled in this extensive guide was sourced from a variety of trusted agricultural universities, research papers, and plant pathology extensions to ensure accuracy and reliability.

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8. Key Takeaways

- **Identify the Bullseye:** Early blight is caused by the *Alternaria solani* fungus. It always starts on the lowest, oldest leaves of the potato plant and creates dark brown spots that feature highly distinct, target-like concentric rings inside them.
- **Moisture is the Enemy:** The fungus requires wet leaves to grow and spread. Keep the leaves dry by using drip irrigation on the ground instead of overhead sprinklers, and give the plants plenty of space so the wind can quickly dry off morning dew.
- **Plant Resistant Varieties:** The easiest way to avoid the disease is to plant potato types that naturally fight it off. Varieties like Sarpo Mira, Defender, Blackberry, Orla, and Setanta are highly resistant and will save you time and money.
- **Rotate Your Crops:** Never plant potatoes in the exact same spot year after year. The fungus sleeps in the dead roots over the winter. Plant your potatoes in a new spot that is at least 225 yards away, and wait 3 to 4 years before returning to the original spot.
- **Clean Up the Garden:** Remove all dead vines at the end of the year and destroy them. Do not compost them unless your compost pile reaches 130 degrees Fahrenheit, or the spores will survive. Pull weeds like hairy nightshade, which hide the disease.
- **Use Natural Shields:** Organic biopesticides made from good bacteria (like *Bacillus subtilis*), copper, baking soda derivatives, or natural oils can safely stop the fungus from attacking the leaves if applied early.
- **Keep Plants Fed:** The fungus attacks weak, starving plants. Ensure your soil has plenty of nitrogen right when the potato plants begin to bloom to keep their natural immune system strong.

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